Vinod Raman

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https://vinodkraman.github.io

Education

University of Michigan — Ann Arbor, MI
PhD Statistics — Sep. 2021 - Dec. 2025
Thesis Advisor: Ambuj Tewari

University of Michigan — Ann Arbor, MI
BSE Computer Science, BSE Chemical Engineering — Sep. 2015 - May 2020
Thesis Advisors: Mahdi Cheraghchi, Sindhu Kutty, Andrej Lenert

Experience

Core Automation — San Francisco, CA
Member of Technical Staff — Sept. 2026 - Present

- Model architecture research.

Google DeepMind — New York, NY
Research Scientist, GenAI — Dec. 2025 - Aug. 2026

- Landed constrained-decoding-based generative retrieval for ads keyword targeting into production.
- Designed RL environments to improve Gemini's capabilities for building end-to-end retrieval systems.
- Used RL to steer models towards generating high-quality and diverse synthetic data.

Google Research — New York, NY
Research Intern, Student Researcher — June - Nov. 2025
Hosts: Matthew Joseph, Travis Dick, Umar Syed

- Designed and implemented new state-of-the-art private algorithms for domain discovery, top-k selection, and submodular maximization over unbounded domains. Paper accepted as an Oral at ICLR 2026.
- Developed an evaluation framework for arithmetic reasoning in frontier LLMs. Paper in submission.
- Built a scalable synthetic data generation pipeline for LLM training, combining fine-tuning and filtering to produce high-quality data at scale.

Apple — New York, NY
Research Intern — Feb. - June 2025
Host: Satyen Kale

- Designed and implemented adaptive allocation strategy for Best-of-N alignment for batch LLM inference.
- Proposed efficient methods for sample-dependent mass estimation. Paper in progress.

Apple — Cupertino, CA
Research Intern — May - Aug. 2024
Host: Kunal Talwar

- Developed state-of-the-art differentially private algorithms for adversarial bandits and online learning for dynamic regret minimization.
- Research published in two papers at ICML 2025.

Awards & Fellowships

- Outstanding Dissertation Award, 2026
- ALT Outstanding Paper Award, 2025
- Apple Scholars in AI/ML PhD Fellowship, 2025
- Outstanding First-Year Ph.D. Student, 2022
- NSF Graduate Research Fellowship, 2022

Publications

(*) denotes equal contribution. $(\alpha$-$\beta)$ denotes alphabetical ordering

24. J. Li, **V. Raman**, A. Tewari. Generation in Metric Spaces.
Conference on Neural Information Processing Systems (NeurIPS), 2026.
https://arxiv.org/abs/2602.07710

23. J. Leng, S. Si, **V. Raman**, H. Yu, I. Dhillon. GroupDPO: Memory efficient Group-wise Direct Preference Optimization.
Conference on Empirical Methods in Natural Language Processing (EMNLP) Findings, 2026.
https://arxiv.org/abs/2604.15602

22. $(\alpha$-$\beta)$ T. Dick, M. Joseph, **V. Raman**. Missing Mass for Differentially Private Domain Discovery.
International Conference on Learning Representations (ICLR), 2026. Oral Presentation.
https://arxiv.org/abs/2603.14016

21. S. Somerstep, **V. Raman***, U. Subedi*, Y. Sun. Learning to Choose or Choosing to Learn: Best-of-N vs. Supervised Fine-Tuning for Bit String Generation.
International Conference on Artificial Intelligence and Statistics (AISTATS), 2026.
https://www.arxiv.org/abs/2505.17288

20. J. Li*, **V. Raman***, A. Tewari. Generation through the lens of learning theory.
Conference on Learning Theory (COLT), 2025.
https://arxiv.org/abs/2410.13714

19. A. Raman*, **V. Raman***. Generation from Noisy Examples.
International Conference on Machine Learning (ICML), 2025.
https://arxiv.org/abs/2501.04179

18. C. Peale*, **V. Raman***, O. Reingold*. Representative Language Generation.
International Conference on Machine Learning (ICML), 2025.
https://arxiv.org/abs/2505.21819

17. H. Asi*, **V. Raman***, A. Saha*. Tracking the Best Expert Privately.
International Conference on Machine Learning (ICML), 2025.
https://arxiv.org/abs/2503.09889

16. **V. Raman**, H. Asi, K. Talwar. Faster Rates for Private Adversarial Bandits.
International Conference on Machine Learning (ICML), 2025.
https://arxiv.org/abs/2505.21790

15. **V. Raman***, U.Subedi*, A.Tewari. The Complexity of Sequential Prediction in Dynamical Systems.
Conference on Learning for Dynamics and Control (L4DC), 2025. Oral Presentation.
https://arxiv.org/abs/2402.06614

14. **V. Raman***, U.Subedi*, A.Tewari. A Unified Theory of Supervised Online Learnability.
Conference on Algorithmic Learning Theory (ALT), 2025. Outstanding Paper Award.
https://arxiv.org/abs/2307.03816

13. **V. Raman***, U.Subedi*, A.Tewari. A Characterization of Multioutput Learnability.
Journal of Machine Learning Research (JMLR), 2024.
https://arxiv.org/abs/2301.02729

12. **V. Raman**, A.Tewari. Online Classification with Predictions.
Conference on Neural Information Processing Systems (NeurIPS), 2024.
https://arxiv.org/abs/2405.14066

11. S.Hanneke*, **V. Raman***, A. Shaeiri*, U.Subedi*. Multiclass Transductive Online Learning.
Conference on Neural Information Processing Systems (NeurIPS), 2024. Spotlight.
https://arxiv.org/abs/2411.01634

10. **V. Raman***, U.Subedi*, A.Tewari. Smoothed Online Classification can be Harder than Batch Classification.
Conference on Neural Information Processing Systems (NeurIPS), 2024.
https://arxiv.org/pdf/2405.15424

9. **V. Raman***, U.Subedi*, A. Raman, A.Tewari. Apple Tasting: Combinatorial Dimensions and Minimax Rates.
Conference on Learning Theory (COLT), 2024.
https://arxiv.org/abs/2310.19064

8. **V. Raman***, U.Subedi*, A.Tewari. Online Learning with Set-Valued Feedback.
Conference on Learning Theory (COLT), 2024.
https://arxiv.org/abs/2306.06247

7. **V. Raman***, U.Subedi*, A.Tewari. Online Infinite-Dimensional Regression: Learning Linear Operators.
Conference on Algorithmic Learning Theory (ALT) 2024.
https://arxiv.org/abs/2309.06548

6. A.Raman, **V. Raman***, U.Subedi*, I.Mehalel*, A.Tewari. Multiclass Online Learnability under Bandit Feedback.
Conference on Algorithmic Learning Theory (ALT) 2024.
https://arxiv.org/abs/2308.04620

5. **V. Raman***, U.Subedi*, A.Tewari. On Proper Learnability between Average- and Worst-case Robustness.
Conference on Neural Information Processing Systems (NeurIPS), 2023.
https://arxiv.org/abs/2211.05656

4. **V. Raman***, U.Subedi*, A.Tewari. On the Learnability of Multilabel Ranking.
Conference on Neural Information Processing Systems (NeurIPS), 2023. Spotlight.
https://arxiv.org/abs/2304.03337

3. (α-β) S.Hanneke, S.Moran, **V. Raman**, U.Subedi, A.Tewari. Multiclass Online Learning and Uniform Convergence.
Conference on Learning Theory (COLT), 2023.
https://arxiv.org/abs/2303.17716

2. **V. Raman**, A.Tewari. Online Agnostic Multiclass Boosting.
Conference on Neural Information Processing Systems (NeurIPS), 2022.
https://arxiv.org/abs/2205.15113

1. **V. Raman**, T.Burger, A.Lenert. Design of thermophotovoltaics for tolerance of parasitic absorption.
Optics Express, 27(22):31757–31772, 2019.
https://doi.org/10.1364/OE.27.031757

In Submission

1. (α-β)V. Feldman, S. Kale, **V. Raman**, K. Talwar, A. Tewari. A Unified Framework for Data-Dependent Probability Estimation. *In Submission*, 2026.

Preprints

4. Y. Kalayci*, **V. Raman***, S. Dughmi. Optimal Stopping vs. Best-of-N for Inference-time Optimization.
In Submission, 2026.
https://arxiv.org/abs/2510.01394

3. S. Xie, **V. Raman**, S. Zhou. Transductive and Learning-Augmented Online Regression.
   Preprint, 2025.

2. **V. Raman**, H. Asi, S. Kale. AdaBoN: Adaptive Best-of-N Alignment.
   Preprint, 2025.
   `https://arxiv.org/abs/2505.12050`

1. **V. Raman***, D.Zhang*, Y.Jung, A.Tewari. Online Boosting for Multilabel Ranking with Top-k Feedback.
   Preprint, 2020.
   `https://arxiv.org/abs/1910.10937`

Selected Talks

7. Optimal Stopping vs. Best-of-N for Inference-time Optimization. *Google DeepMind Tech Talk*, 2025.

6. A Unified Theory of Supervised Online Learning. *ALT*, 2025.

5. Generation through the lens of learning theory. *Apple MLR Reading Group*, 2025.

4. Generation through the lens of learning theory. *NEU CS Theory Seminar*, 2024.

3. Trichotomies in Online Learnability. *Apple MLR Reading Group*, 2024.

2. Multiclass Online Learning and Uniform Convergence. *UMich EECS Theory Seminar*, 2024.

1. On Classification-Calibration of Gamma-Phi Losses. *COLT*, 2023.

References

1. **Ambuj Tewari**, Professor, Statistics, University of Michigan, Ann Arbor MI, USA.
   Email: tewaria@umich.edu | *Phone*: 734-615-0928

2. **Steve Hanneke**, Assistant Professor, Computer Science, Purdue University, West Lafayette IN, USA.
   Email: steve.hanneke@gmail.com

3. **Mahdi Cheraghchi**, Associate Professor, Computer Science, University of Michigan, Ann Arbor MI, USA.
   Email: mahdich@umich.edu | *Phone*: 734-763-9165